

Seminar 1. (1.10.2025)

1. ADDITION
2. MULTIPLICATION
3. DIVISION
4. SUBTRACTION

1. ADDITION

a)
$$\begin{array}{r} 11 \\ 3254(7) \\ 616(7) \\ \hline 4203 \end{array} +$$

$4+6=10 \begin{cases} 17=1 \\ \%7=3 \end{cases}$

$5+1+1=7 \begin{cases} 17=1 \\ \%7=0 \end{cases}$
 $6+2+1=9 \begin{cases} 17=1 \\ \%7=2 \end{cases}$

$3+1=4 < 7$

b)
$$\begin{array}{r} 11 \\ 739(10) \\ 85(10) \\ \hline 824(10) \end{array} +$$

$9+5=14 / 10 = 1$
 $\%10 = 4$

$8+3+1=12 / 10 = 1$
 $\%10 = 2$

$7+1=8 < 10$

A=10

B=11

C=12

D=13

E=14

F=15

c)
$$\begin{array}{r} 111111 \\ 10100111(2) \\ 11101(2) \\ \hline 11000100(2) \end{array} +$$

$1+1=2 / 2 = 1$
 $\%2 = 0$

d)
$$\begin{array}{r} 11 \\ AFF92(16) \\ B4D(16) \\ \hline B0ADF(16) \end{array} +$$

$D+2=13+2=15=F < 16$

$9+4=13=D$

$B+F=11+15=26 / 16 = 1$
 $\%16 = 10 = A$

$F+1=16 / 16 = 1$
 $\%16 = 0$

$A+1=11=B$

2. MULTIPLICATION

a)
$$\begin{array}{r} 2323 \\ 3546(8) \\ \cdot \\ 5(8) \\ \hline 22376(8) \end{array} \cdot$$

$6 \cdot 5 = 30 / 8 = 3$
 $\%8 = 6$

$2+5 \cdot 5 = 27 / 8 = 3$
 $\%8 = 3$

$3+4 \cdot 5 = 23 / 8 = 2$
 $\%8 = 7$

$3+3 \cdot 5 = 18 / 8 = 2$
 $\%8 = 2$

b)
$$\begin{array}{r} 212 \\ 4035(6) \\ \cdot \\ 3(6) \\ \hline 20153(6) \end{array} \cdot$$

$5 \cdot 3 = 15 / 6 = 2$
 $\%6 = 3$

$1+0 = 1 / 6 = 0$
 $1 \%6 = 1$

$3 \cdot 3 + 2 = 11 / 6 = 1$

$4 \cdot 3 = 12 / 6 = 2$

$$\%6 = 5$$

$$\%6 = 0$$

$$\begin{array}{r} 9 \ 6 \ 3 \ 5 \\ c) \ \overline{FA581(16)} \\ \quad A(16) \\ \hline 9 \ C \ 7 \ 7 \ 0 \ A \end{array} .$$

$$A \cdot 1 = 10 / 16 = 0$$

$$\%16 = 10 = A$$

$$8 \cdot A = 80 / 16 = 5$$

$$\%16 = 0$$

$$5 + 50 = 55 / 16 = 3$$

$$\%16 = 7$$

$$3 + 100 = 103 / 16 = 6$$

$$\%16 = 7$$

$$9 + 150 = 159 / 16 = 9$$

$$\%16 = 12 = C$$

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3. Division

$$\begin{array}{r} a) \ \overline{23151(8)} \mid \overline{5(8)} \\ \quad \underline{23} \\ \quad \underline{41} \\ \quad \underline{35} \\ \quad \underline{41} \\ \quad \underline{3} \end{array}$$

$$23(8) = 3 \cdot 8^0 + 2 \cdot 8^1 = 3 + 16 = 19(10) / 5 = 3$$

$$\%5 = 4$$

$$41(8) = 1 \cdot 8^0 + 4 \cdot 8^1 = 1 + 32 = 33 / 5 = 6$$

$$\%5 = 3$$

$$35(8) = 5 \cdot 8^0 + 3 \cdot 8^1 = 5 + 24 = 29 / 5 = 5$$

$$\%5 = 4$$

$$\begin{array}{r} b) \ \overline{23151(10)} \mid \overline{5(10)} \\ \quad \underline{23} \\ \quad \underline{31} \\ \quad \underline{15} \\ \quad \underline{01} \end{array}$$

$$\begin{array}{r} c) \ \overline{4654(8)} \mid \overline{5(8)} \\ \quad \underline{46} \\ \quad \underline{35} \\ \quad \underline{44} \\ \quad \underline{1} \end{array}$$

$$46(8) = 6 + 32 = 38 / 5 = 7$$

$$\%5 = 3$$

$$35(8) = 5 + 24 = 29 / 5 = 5$$

$$\%5 = 4$$

$$44(8) = 4 + 32 = 36 / 5 = 7$$

$$\%5 = 1$$

$$\begin{array}{r} d) \ \overline{2154(6)} \mid \overline{3(6)} \\ \quad \underline{21} \\ \quad \underline{04} \\ \quad \underline{35} \end{array}$$

$$21(6) = 1 + 12 = 13 / 3 = 4$$

$$\%3 = 1$$

$$15(6) = 5 + 6 = 11 / 3 = 3$$

$$\begin{array}{r} 15 \\ / \\ 24 \\ / \\ 1 \end{array}$$

$$\begin{aligned} 15(6) &= 5 + 6 = 11 / 3 = 3 \\ \%3 &= 2 \\ 24(6) &= 4 + 12 = 16 / 3 = 5 \\ \%3 &= 1 \end{aligned}$$

$$\begin{array}{r} \text{e) } BF8D(16) \mid B(16) = 11(16) \\ \underline{B} \\ F \\ / \\ 48 \\ / \\ 6D \\ / \\ A \end{array}$$

$$\begin{aligned} F(16) &= 15 / 11 = 1 \\ \%11 &= 4 \end{aligned}$$

$$48(16) = 8 \cdot 16^0 + 4 \cdot 16 = 8 + 64 = 72 / 11 = 6$$

$$\%11 = 6$$

$$6D(16) = 13 + 6 \cdot 16 = 96 + 13 = 109 / 11 = 9$$

$$\%11 = 10$$

$$\begin{array}{r} \text{f) } 7312(8) \mid 5(8) \\ \underline{5} \\ 23 \\ / \\ 41 \\ / \\ 32 \\ / \\ 1 \end{array}$$

$$\begin{aligned} 23(8) &= 19 / 5 = 3 \\ \%5 &= 4 \end{aligned}$$

$$\begin{aligned} 41(8) &= 33 / 5 = 6 \\ \%5 &= 3 \end{aligned}$$

$$\begin{aligned} 32(8) &= 26 / 5 = 5 \\ \%5 &= 1 \end{aligned}$$

4. SUBTRACTION

$$\begin{array}{r} \text{a) } 3342(5) \\ 3144(5) \\ \hline 143(5) \end{array}$$

$$\begin{aligned} 2 < 4 &\Rightarrow \text{borrow the base (here, 5)} \Rightarrow 2 - 4 + 5 = \underline{3} \\ 4 - 4 - 1 + 5 &= \underline{4} \\ 3 - 1 - 1 &= \underline{1} \end{aligned}$$

$$\begin{array}{r} \text{b) } 1010100(2) \\ 11111(2) \\ \hline 1000101(2) \end{array}$$

$$\begin{aligned} 0 < 1 &\Rightarrow \text{borrow} \Rightarrow 0 - 1 + 2 = 1 \\ 0 < 1 &\Rightarrow \dots \Rightarrow 0 - 1 - 1 + 2 = 0 \\ 1 - 1 - 1 + 2 &= 1 \\ 0 - 1 - 1 + 2 &= 0 \end{aligned}$$

$$\begin{array}{r} \text{c) } E4AB9(16) \\ 6AFD(16) \\ \hline DDFBC(16) \end{array}$$

$$\begin{aligned} 9 < D &\Rightarrow \text{borrow} \Rightarrow 9 - 13 + 16 = 12 = C \\ 11 - 15 - 1 + 16 &= 11 = B \\ 10 - 10 - 1 + 16 &= 15 = F \\ 4 - 6 - 1 + 16 &= 13 = D \\ 14 - 1 &= 13 = D \end{aligned}$$